

Binary Depression Classifier

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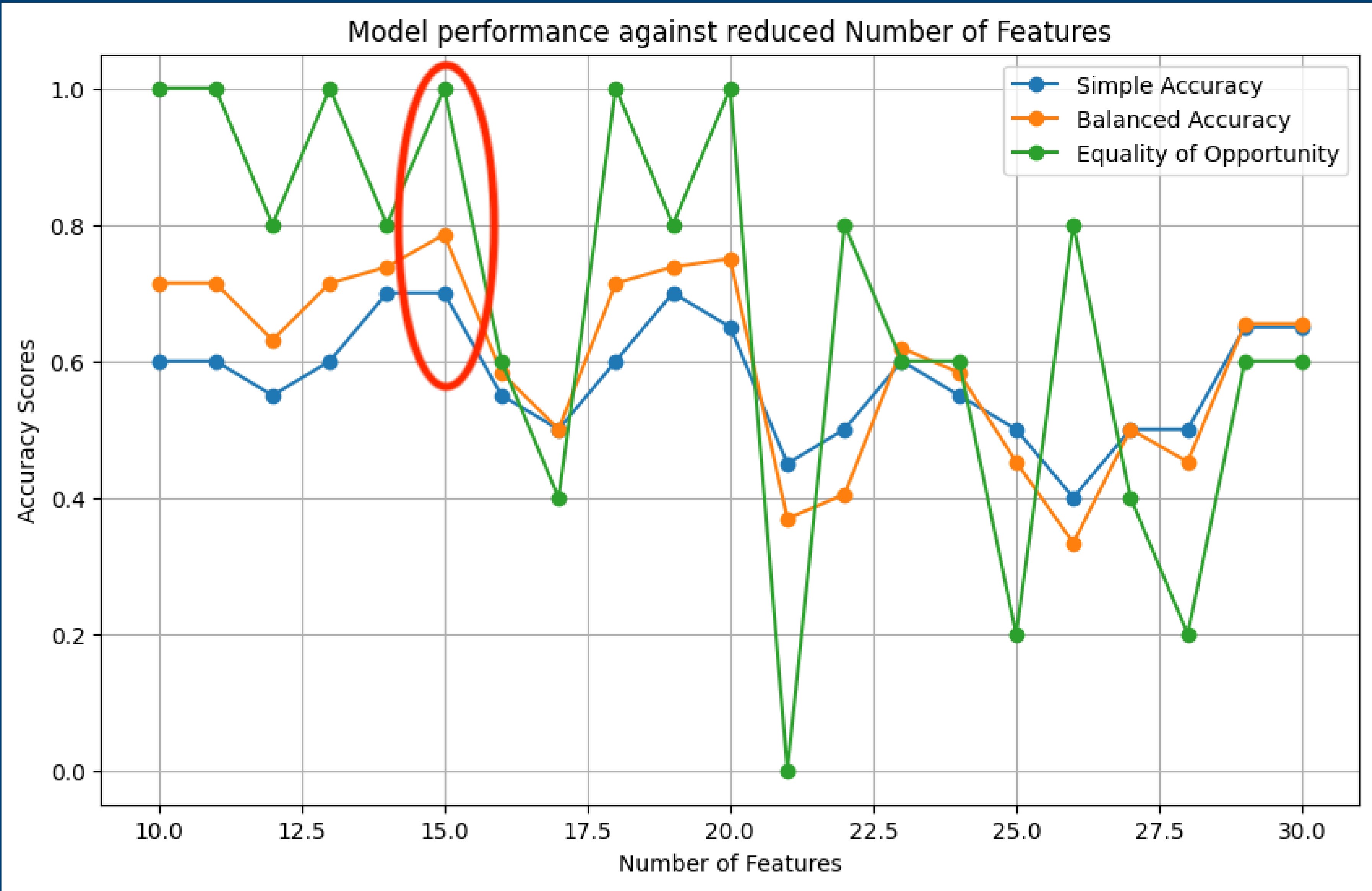
Order of Tasks

- Import, Normalize, and Clean up Data
- Use SMOTE for imbalanced Classification
- Use XGBoost for both **Depression** and **Gender** Classification
- Find most informative features for both **Depression** and **Gender** using χ^2 test
- Mitigate gender bias by removing features highly indicative of gender from depression classification

Why use SMOTE?

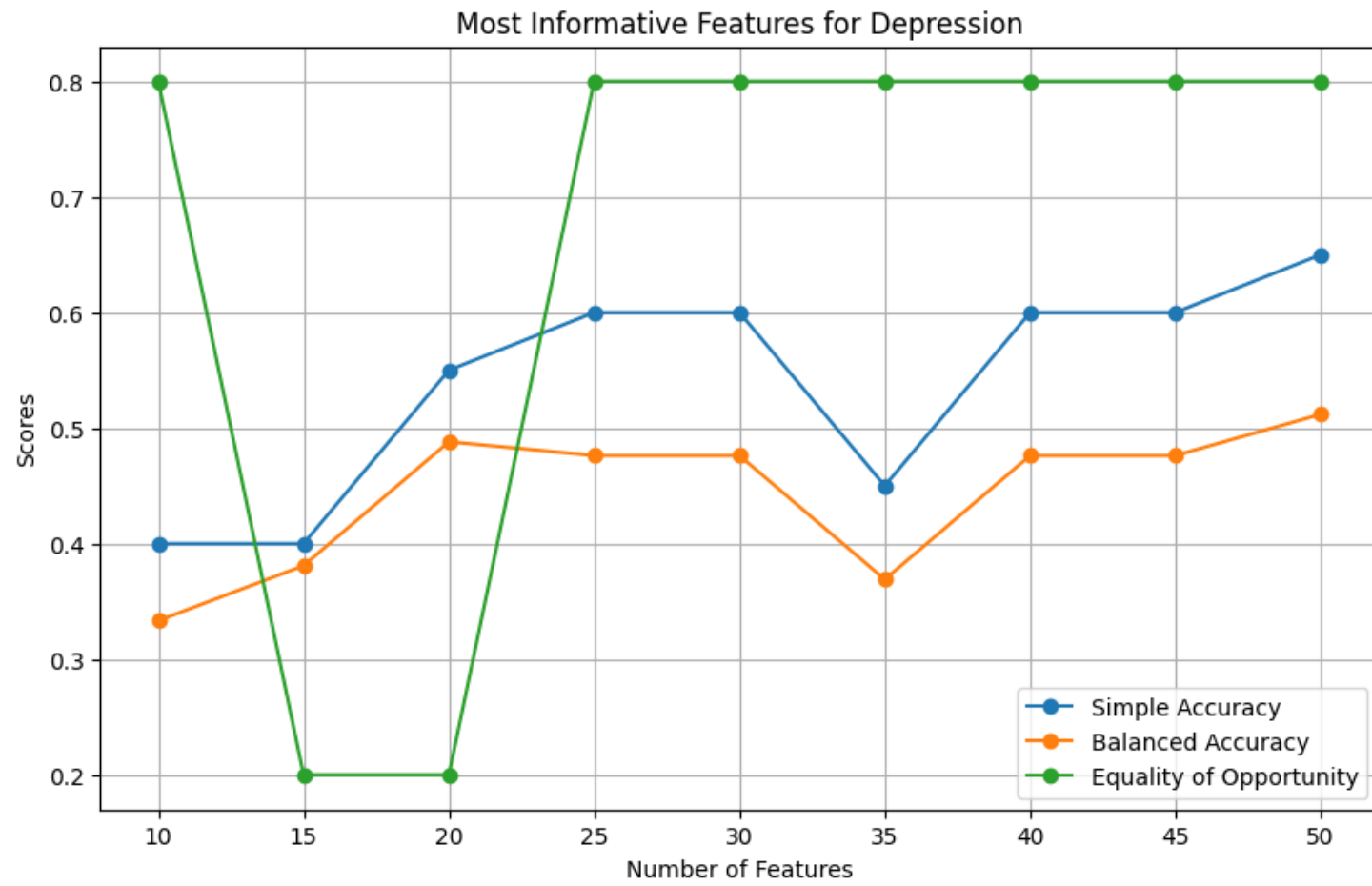
- SMOTE is an oversampling algorithm
- It picks out under-represented samples using the underlying distribution to equalize the class sizes
- Our algorithm was biased towards classifying all test participants as **NOT** depressed
- Gender classification using XGBoost provided almost 40% of the importance to one features
- Depression classification using the same algorithm had the highest importance for one feature at 4%

XGboost is capable of significantly improving both balanced accuracy and equality of opportunity given we remove features dependent upon gender.



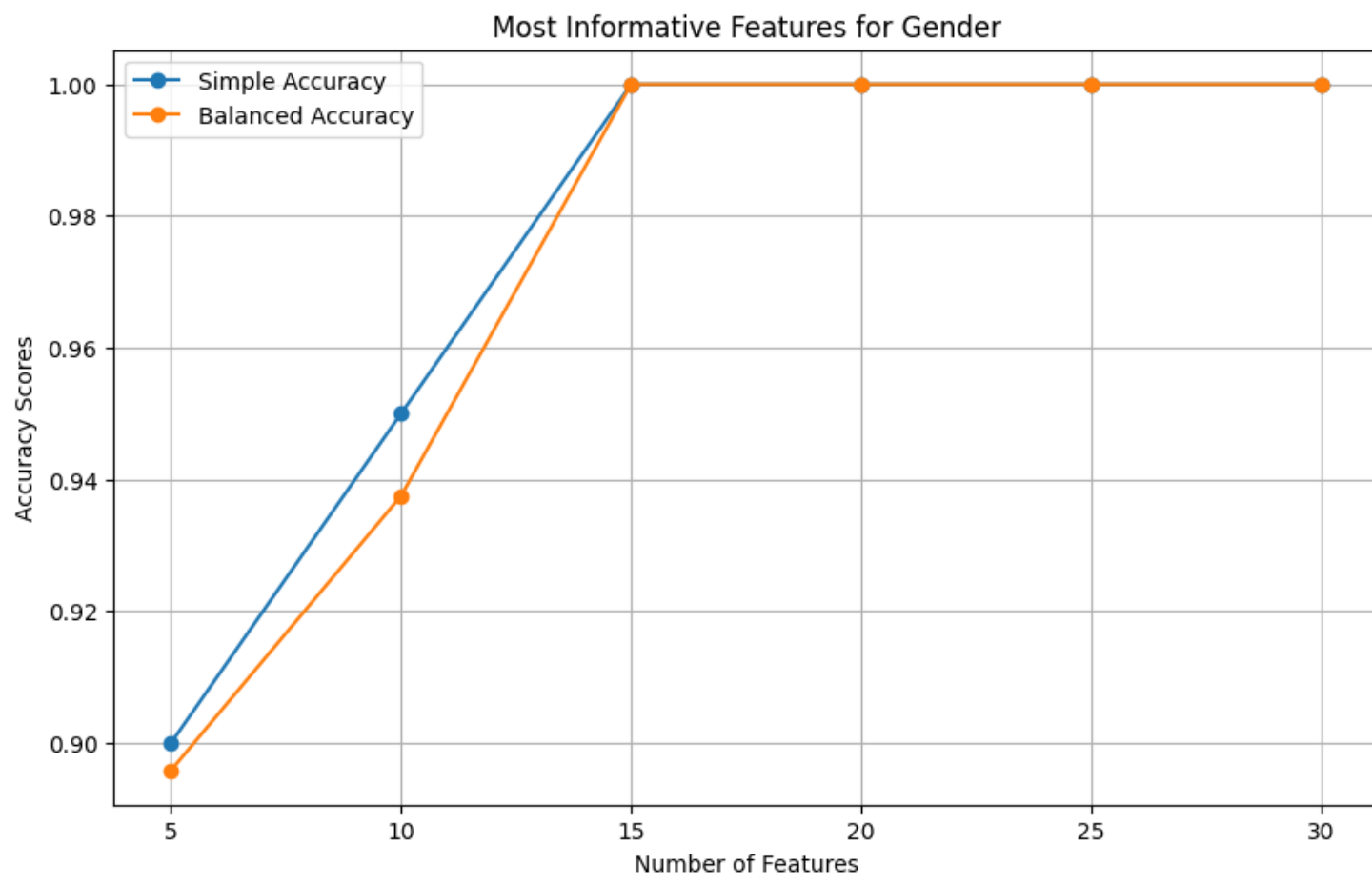
Most Important Features for Depression

loudness_sma3_amean
loudness_sma3_percentile50.0
loudness_sma3_percentile80.0
loudness_sma3_pctlrange0-2
loudness_sma3_meanRisingSlope
loudness_sma3_stddevRisingSlope
loudness_sma3_stddevFallingSlope
spectralFlux_sma3_amean
HNRdBACF_sma3nz_amean
spectralFluxV_sma3nz_amean



Most Important Features for Gender

F0semitoneFrom27.5Hz_sma3nz_amean
F0semitoneFrom27.5Hz_sma3nz_percentile20.0
F0semitoneFrom27.5Hz_sma3nz_percentile50.0
F0semitoneFrom27.5Hz_sma3nz_percentile80.0
jitterLocal_sma3nz_stddevNorm
shimmerLocaldB_sma3nz_stddevNorm
HNRdBACF_sma3nz_amean
F1frequency_sma3nz_stddevNorm
slopeV0-500_sma3nz_amean
mfcc4V_sma3nz_amean



Impact of Feature Removal

Evaluation Metrics	XGBoost	
	Before Feature Elimination	After Feature Elimination
Accuracy	70%	70%
Balanced Accuracy	50%	79%
Equality of Opportunity	100%	100%